Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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RH_picture3.png

RH_banner.png

NavyBlue **NanoJ: High-Performance Open-Source Super-Resolution Microscopy in ImageJ** Black

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Our team has built an open-source image analysis framework for super-resolution microscopy designed to combine high performance and ease of use. We named it NanoJ - a reference to the popular ImageJ software it was developed for. Here we highlight the current capabilities of NanoJ for several essential processing steps: spatio-temporal alignment of raw data (NanoJ-Core), super-resolution image reconstruction (NanoJ-SRRF), image quality assessment (NanoJ-SQUIRREL), structural modelling (NanoJ-VirusMapper) and control of the sample environment (NanoJ-Fluidics). We expect to expand NanoJ in the future through the development of new tools designed to improve quantitative data analysis and measure the reliability of fluorescent microscopy studies.

FigureMain_v4.png



NanoJ-Core: Drift Correction

Drift estimation is achieved by directly analysing a raw sequence of timepoints. Frames can then be directly translated to correct drift or a table with drift coordinates can be generated and used in other algorithms (e.g. SRRF).

FigureDrift_v2.png



figureDrift correction with NanoJ-Core.

a) Composite image of two frames from a time-lapse dataset of the same field-of-view. **b)** Cross-correlation map (CCM) between the two frames shown in a). The vector position of the maximum indicates the linear shift between the two frames. **c)** Overlay of the two frames after drift correction using NanoJ-Core. **d)** Vertical and horizontal drift curves obtained using NanoJ-Core from the 100-frame raw data.

NanoJ-Core: Channel Registration

Channel registration is able to calculate and apply an elastic transform to realign different captured wavelengths.

FigureChannelAlignment_v4.png


FigureChannelAlignment_v4.png

figure**Multi-colour channel registration**

with NanoJ-Core. a) Composite image of multi-colour beads, prior to (left) and after (right) channel registration using NanoJ-Core. Scale bars: 25 µm, insets: 0.5 µm. **b)** Vectorial representation of the shift between the two channels (left, displacement vector length 50 times larger for representation purposes), horizontal (middle) and vertical (right) shift maps obtained and applied to the data shown in a). Scale bars: 25 µm.

NanoJ-SRRF: Live-Cell Super-Resolution

Super-Resolution Radial Fluctuations (SRRF) is able to generate a super-resolution reconstruction by analysing fluctuations in the emission of fluorophores captured in a short image sequence.

FigureSRRF_v6.png

figure**Live-cell super-resolution**

microscopy with NanoJ-SRRF. a) Comparison of widefield and SRRF reconstruction from UtrCH-GFP actin labelling. Scale bar: 5 µm. **b)** Time-course of the inset shown in a), obtained at 33.3 Hz and displayed every 30 s. Scale bar: 1 µm. **c)** Colour-coded time course. Scale bar: 1 µm.

NanoJ-SQUIRREL: Estimating Image Quality & Resolution

SQUIRREL quantitatively calculates local quality and resolution in super-resolution images. It highlights limitations on the data collected and a reference point to help researchers improve imaging fidelity.

FigureSQUIRREL_v3.png

figure**Quality and resolution**

assessment with NanoJ-SQUIRREL. a) A super-resolution rendering and acquired widefield image of fixed Alexa647 labelled microtubules. **b)** Left: SQUIRREL error map highlighting discrepancies between the super-resolution and diffraction-limited images in (a). Right: Magnified insets at indicated positions on error map. **c)** Left: SQUIRREL resolution map of the super-resolution image in (a). Right: Magnified insets for indicated resolution blocks. Whole image scale bars = 5 µm, inset scale bars = 1 µm.

NanoJ-VirusMapper: Structural Mapping and Modelling

VirusMapper features a single-particle analysis (SPA) algorithm, capable of generating structural models of conserved structures imaged by Super-Resolution Microscopy

Figure_VirusMapper.png

figure**Quantitative SPA-based modelling.** Multicomponent model of the Vaccinia virus by imaging in super-resolution hundreds of fluorescently labelled viruses and modelling their structure through VirusMapper

NanoJ-Fluidics: Liquid Exchange

NanoJ-Fluidics (Pumpy McPumpface) entails the control of a simple inexpensive LEGO-based syringe pump array that automates live-to-fixed imaging and sequential labelling of the sample. Protocols can be run automatically in parallel to the acquisition.

Pumpy_Figure1_NComm_Rev3_v6.png

figure**Schematics of the NanoJ-Fluidics system. a)** 3D side view of a single syringe pump. **b)** 2D top view of a syringe pump array (representing 4 pumps out of 128 maximum) and a fluid extraction peristaltic pump, both controlled by an Arduino UNO. **c)** Example of possible workflows

Author representation



We recognise a gender inequality in the author list (2F/8M). I am committed to achieve a better balanced representation of gender and minorities in our future work.

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